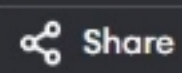


main.c



Run

Output

main.c

```
printf("Enter the key: ");
scanf("%s", key);

int keyLength = strlen(key);

for (i = 0; plaintext[i] != '\0'; i++) {
    if (isalpha(plaintext[i])) {
        int shift = toupper(key[j % keyLength]) - 'A';

        if (isupper(plaintext[i]))
            ciphertext[i] = ((plaintext[i] - 'A' + shift) % 26) + 'A';
        else
            ciphertext[i] = ((plaintext[i] - 'a' + shift) % 26) + 'a';

        j++;
    } else {
        ciphertext[i] = plaintext[i];
    }
}

ciphertext[i] = '\0';
```

Enter the plaintext: HELLO
Enter the key: KEY
Ciphertext: RIJVS

=== Code Execution Successful ===

main.c

Share

Run

19

printf("Enter Ciphertext: ");

20

scanf("%s", cipher);

21

22

printf("\nPossible Plaintexts:\n");

23

24

for (i = 0; i < 12; i++) {

25

a = aValues[i];

26

inv = modInverse(a);

27

28

for (b = 0; b < 26; b++) {

29

printf("\na = %d, b = %d : ", a, b);

30

31

for (j = 0; cipher[j] != '\0'; j++) {

32

if (isupper(cipher[j])) {

33

int x = cipher[j] - 'A';

34

int p = (inv * (x - b + 26)) % 26;

35

printf("%c", p + 'A');

36

} else {

37

printf("%c", cipher[j]);

38

}

39

}

40

}

Output

Possible Plaintexts:

a = 1, b = 0 : hello

a = 1, b = 1 : hello

a = 1, b = 2 : hello

a = 1, b = 3 : hello

a = 1, b = 4 : hello

a = 1, b = 5 : hello

a = 1, b = 6 : hello

a = 1, b = 7 : hello

a = 1, b = 8 : hello

a = 1, b = 9 : hello

a = 1, b = 10 : hello

a = 1, b = 11 : hello

a = 1, b = 12 : hello

a = 1, b = 13 : hello

a = 1, b = 14 : hello

a = 1, b = 15 : hello

a = 1, b = 16 : hello

a = 1, b = 17 : hello

a = 1, b = 18 : hello

main.c



Share

Run

Output

Clear

```
4- int main() {  
5-     char text[100];  
6-     int a, b, i;  
7-  
8-     printf("Enter the plaintext: ");  
9-     scanf("%s", text);  
10-  
11-    printf("Enter the value of a: ");  
12-    scanf("%d", &a);  
13-  
14-    printf("Enter the value of b: ");  
15-    scanf("%d", &b);  
16-  
17-    for (i = 0; text[i] != '\0'; i++) {  
18-        if (isupper(text[i])) {  
19-            text[i] = ((a * (text[i] - 'A') + b) % 26) + 'A';  
20-        }  
21-        else if (islower(text[i])) {  
22-            text[i] = ((a * (text[i] - 'a') + b) % 26) + 'a';  
23-        }  
24-    }  
25-}
```

Enter the plaintext: kavi
Enter the value of a: 3
Enter the value of b: 4
Ciphertext: iepc

=== Code Execution Successful ===

```
main.c
34
35 printf("Ciphertext: ");
36
37 for (i = 0; text[i] != '\0'; i += 2) {
38     findPosition(text[i], &r1, &c1);
39     findPosition(text[i + 1], &r2, &c2);
40
41     if (r1 == r2) {
42         printf("%c%c",
43             matrix[r1][(c1 + 1) % 5],
44             matrix[r2][(c2 + 1) % 5]);
45     }
46     else if (c1 == c2) {
47         printf("%c%c",
48             matrix[(r1 + 1) % 5][c1],
49             matrix[(r2 + 1) % 5][c2]);
50     }
51     else {
52         printf("%c%c",
53             matrix[r1][c2],
54             matrix[r2][c1]);
55     }
56 }
```

Output

Enter plaintext (uppercase, even letters): HELLOWORLD
Ciphertext: CFPPNVNMTC

=== Code Execution Successful ===



main.c



Share

Run

Output

C

```
1  #include <stdio.h>
2  #include <string.h>
3  #include <ctype.h>
4
5  int main() {
6      char plaintext[100];
7      char key[] = "QWERTYUIOPASDFGHJKLZXCVBNM";
8      int i;
9
10     printf("Enter the plaintext: ");
11     fgets(plaintext, sizeof(plaintext), stdin);
12
13     for (i = 0; plaintext[i] != '\0'; i++) {
14         if (isupper(plaintext[i])) {
15             plaintext[i] = key[plaintext[i] - 'A'];
16         }
17         else if (islower(plaintext[i])) {
18             plaintext[i] = tolower(key[plaintext[i] - 'a']);
19         }
20     }
21 }
```

Enter the plaintext: kavipriya
Ciphertext: aqcohkong

=== Code Execution Successful ===

```
main.c
38
39 printf("Ciphertext: ");
40
41 for (i = 0; text[i] != '\0'; i += 2) {
42     findPos(text[i], &r1, &c1);
43     findPos(text[i + 1], &r2, &c2);
44
45     if (r1 == r2) {
46         printf("%c%c",
47             matrix[r1][(c1 + 1) % 5],
48             matrix[r2][(c2 + 1) % 5]);
49     }
50     else if (c1 == c2) {
51         printf("%c%c",
52             matrix[(r1 + 1) % 5][c1],
53             matrix[(r2 + 1) % 5][c2]);
54     }
55     else {
56         printf("%c%c",
57             matrix[r1][c2],
58             matrix[r2][c1]);
59     }
```

Output

Enter plaintext (uppercase, no spaces, even length): ppercase, no spaces, even length):
MUSTSEYOUOVERCADOGANWESTCOMINGATONCE
Ciphertext: RQTXL...
Ciphertext: HMMHMMHMMHMM

=== Code Execution Successful ===

-
-
-
-
-
-
-
-
-
-
-
-

main.c

Run

Output

Clear

```
5- int main() {
6-     char plain[100], cipher[100];
7-     char key[] = "CIPHERABDFGJKLMNOPQRSTUVWXYZ";
8-     int i;
9-
10-    printf("Enter the plaintext: ");
11-    fgets(plain, sizeof(plain), stdin);
12-
13-    for (i = 0; plain[i] != '\0'; i++) {
14-        if (isupper(plain[i])) {
15-            cipher[i] = key[plain[i] - 'A'];
16-        }
17-        else if (islower(plain[i])) {
18-            cipher[i] = tolower(key[plain[i] - 'a']);
19-        }
20-        else {
21-            cipher[i] = plain[i];
22-        }
23-    }
24-
25-    cipher[i] = '\0';
26-}
```

Enter the plaintext: HELLO
Ciphertext: BEJJM

=== Code Execution Successful ===

Programiz

C Online Compiler

Programiz PRO >

main.c

Run

Share

1

#include <stdio.h>

2

#include <string.h>

3

4

int main() {

5

char cipher[1000];

6

int freq[256] = {0};

7

int i;

8

9

printf("Enter the ciphertext:\n");

10

fgets(cipher, sizeof(cipher), stdin);

11

12

for (i = 0; cipher[i] != '\0'; i++) {

13

freq[(unsigned char)cipher[i]]++;

14

}

15

16

printf("\nCharacter Frequencies:\n");

17

for (i = 0; i < 256; i++) {

18

if (freq[i] > 0 && i != '\n')

19

printf("%c : %d\n", i, freq[i]);

20

}

21

22

return 0;

Output

Clear

Enter the ciphertext:

53444305))6*;4826...

Character Frequencies:

) : 2

* : 1

. : 3

0 : 1

2 : 1

3 : 2

4 : 1

5 : 2

6 : 2

8 : 1

; : 1

: 3

: 1

: 2

: 3

=== Code Execution Successful ===